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 Studierichting: Informatica
 Oefeningenreeks 5D nr. 3.3

$$f(x) = x^2 + 4x \text{ en } g(x) = x - 2$$

$$v(x) = f(x) - g(x)$$

$$v(x) = x^2 + 3x + 2$$

$$\Rightarrow D = 9 - 4 \cdot 1 \cdot 2 = 1$$

$$x_1 = \frac{-3+1}{2} = -1$$

$$x_2 = \frac{-3-1}{2} = -2$$

x	-2	-1
$v(x)$	$+$	$+$

$$\begin{aligned}
 I &= \int_{-3}^0 (x^2 + 3x + 2) dx \\
 &= \int_{-3}^{-2} (x^2 + 3x + 2) dx - \int_{-2}^{-1} (x^2 + 3x + 2) dx + \int_{-1}^0 (x^2 + 3x + 2) dx \\
 &= \left[\frac{x^3}{3} + \frac{3x^2}{2} + 2x \right]_{-3}^{-2} - \left[\frac{x^3}{3} + \frac{3x^2}{2} + 2x \right]_{-2}^{-1} + \left[\frac{x^3}{3} + \frac{3x^2}{2} + 2x \right]_{-1}^0 \\
 &= \left(\frac{-8}{3} + 6 - 4 \right) - \left(\frac{-27}{3} + \frac{27}{2} - 6 \right) - \left[\left(\frac{-1}{3} + \frac{3}{2} - 2 \right) - \left(\frac{-8}{3} + 6 - 4 \right) \right] - \\
 &\quad \left(\frac{-1}{3} + \frac{3}{2} - 2 \right) \\
 &= \frac{5}{6} + \frac{1}{6} + \frac{5}{6} = \frac{11}{6}
 \end{aligned}$$

Nodig voor de tekening:

$$f(x) = x^2 + 4x \text{ met nulpunten } x = 0 \text{ en } x = -4$$

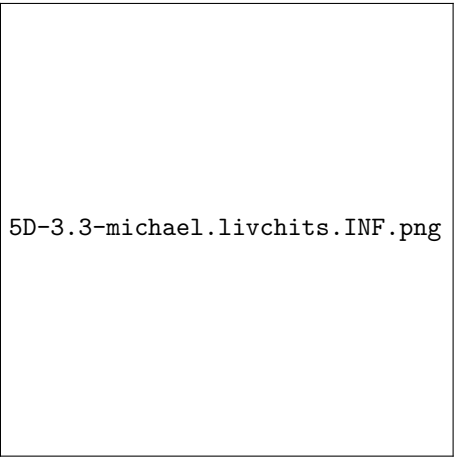
$$f'(x) = 2x + 4 \text{ met nulpunt } x = -2$$

x	-4	-2	0
$f(x)$	$+$	0	$+$
$f'(x)$	$\searrow$	$\searrow$	$\nearrow$
$f''(x)$	$\frown$	$\frown$	$\frown$

$$g(x) = x - 2 \text{ met nulpunt } x = 2$$

$$g'(x) = 1$$

x	2		
$g(x)$	$-$	0	$+$
$g'(x)$	$\nearrow$	$\nearrow$	$\nearrow$



5D-3.3-michael.livchits.INF.png

References